Midterm 2019/2020

Draw Gantt chart + Average Waiting + Average Response

Process	Arrival	Cpu-Burst	Priority
P_1	0	5 2	1
P_2	3	3	0
P_3	5	8	3
P_4	7	4 3	3
P_5	9	3	2

1) Non preemptive priority scheduling with SJF

P_1	P_2	P_4	P_5	P_3

0 5 8 12 15 23

Average waiting time = Average Response time

$P_1 = 0 - 0 = 0$ $P_2 = 5 - 3 = 2$, $P_3 = 15 - 5 - 10$, $P_4 = 8 - 7 = 1$,

$P_5 = 12 - 9 = 3$

Average waiting = Average Response $= \frac{16}{5} = 3.2$ units

2) preemptive priority scheduling + Shortest Remaining time first

P_1	P_2	P_1	P_4	P_5	P_4	P_3

0 3 6 8 9 12 15 23

Waiting time = All waiting - No of units processed - Arrival time

$P_1 = 6 - 3 - 0 = 3$, $P_2 = 3 - 3 = 0$, $P_3 = 15 - 5 = 10$

$P_4 = 12 - 1 - 7 = 4$, $P_5 = 9 - 9 = 0$

Average waiting time $= \frac{3 + 10 + 4}{5} = \frac{17}{5} = 3.4$ units

Response time = First time - Arrival

$P_1 = 0 - 0 = 0$, $P_2 = 3 - 3 = 0$, $P_3 = 15 - 5 = 10$, $P_4 = 8 - 7 = 1$,

$P_5 = 9 - 9 = 0$

Average Response $= \frac{11}{5} = 2.2$ units

Midterm 2018

Gantt Chart + Average waiting + Average Response time

Process	Arriving	Burst	Priority
P_1	0	6	1
P_2	3	2	0
P_3	5	3	2
P_4	7	6	4
P_5	9	3	3

[1] Shortest Job First

P_1	P_2	P_3	P_5	P_4
0 – 6	6 – 8	8 – 11	11 – 14	14 – 20

Average waiting = Average Response time

Waiting = All waiting − Arrival

$P_1 = 0$, $P_2 = 6-3 = 3$, $P_3 = 8-5 = 3$, $P_4 = 14-7 = 7$, $P_5 = 11-9 = 2$

Average waiting = Average Response = $\frac{15}{5} = 3$ units

[2] SRJF = Shortest Job First + Preemptive

~~P_1~~	~~P_2~~	~~P_3~~	~~P_5~~	~~P_4~~
~~0 – 6~~	~~6 – 8~~			~~– 20~~

P_1	P_2	P_1	P_3	P_5	P_4
0 – 3	3 – 5	5 – 8	8 – 11	11 – 14	14 – 20

Waiting time = All − no. of units processed − Arrival

$P_1 = 5 - 3 - 0 = 2$

$P_2 = 3 - 0 - 3 = 0$

$P_3 = 8 - \cancel{5} - 0 = 3$

$P_4 = 14 - 7 = 7$

$P_5 = 11 - 9 = 2$

Average waiting = $\frac{14}{5} = 2.8$ units

Average Response =

$P_1 = 0$, $P_2 = 0$, $P_3 = 8 - 5 = 3$, $P_4 = 14 - 7 = 7$, $P_5 = 11 - 9 = 2$

Average Resp = $\frac{12}{5} = 2.4$ units

Final 2019/2020

Question 2

Draw the Gantt chart then Calculate Average waiting time and Average Response time

Lowest integer = highest priority

Process	Arrival time	Burst	Priority
P1	0	8	3
P2	3	4	2
P3	5	4	1
P4	7	2	1
P5	8	2	0

1) First Come first serve

P1	P2	P3	P4	P5

0 8 12 16 18 20

Response time only differs in preemptive Scheduling

Average waiting time = Average Response time

Waiting time = All waiting - Arrival

P_1 waiting = 0 , $P_2 = 8-3 = 5$, $P_3 = 12-5 = 7$

$P_4 = 16-7 = 9$, $P_5 = 18-8 = 10$

Average waiting $= \frac{31}{5} = 6.2$ units = Average Response time

2) Nonpreemptive SJF

P1	P4	P5	P2	P3

0 8 10 12 16 20

Average waiting = ? = Average ~~Arrival~~ Response time

$P_1 = 0$, $P_2 = 12-3 = 9$, $P_3 = 16-5 = 11$, $P_4 = 8-7 = 1$,

$P_5 = 10-8 = 2$

Average $= \frac{23}{5} = 4.6$ units ⇐ better than FCFS

3) Non preemptive with priority

Average waiting = Average Response

P1	P5	P4	P3	P2

0 8 10 12 16 20

$P_1 = 0$, $P_2 = 16-3 = 13$,

$P_3 = 12-5 = 7$, $P_4 = 10-7 = 3$, $P_5 = 8-8 = 0$ ⇒ Average waiting = 4.6 units

[redacted]

1) preemptive shortest Remaining time First with Priority

	Arrival	Cpu Burst	Priority
P_1	0	~~8~~ 5	3
P_2	3	~~4~~	2
P_3	5	4	1
P_4	7	~~2~~	1
P_5	8	~~2~~	0

P_1	~~P_2~~ P_2	P_4	P_5	P_3	P_1

0, 3, 5, 7, 9, 11, 15

Waiting time = All waiting - Number of units processed - Arrival time

P_1 = 15 - 3 - 0 = 12

P_2 = 3 - 0 - 3 = 0

P_3 = 11 - 0 - 5 = 6

P_4 = 7 - 0 - 7 = 0

P_5 = 9 - 0 - 8 = 1

Average waiting = $\frac{18+1}{5}$ = 3.8 units

Response time = First Response - Arrival

P_1 = 0 - 0 = 0

P_2 = 3 - 3 = 0

P_3 = 11 - 5 = 6

P_4 = 7 - 7 = 0

P_5 = 9 - 8 = 1

Average Response = $\frac{7}{5}$ = 1.4 units

2019 Exam Final

Average time = Average Seek time + Average latency + transfer time + Controller over head

Average seek time = 14 ms

Average latency = $\frac{1}{2} *$ latency = $\frac{1}{2} * \frac{60}{7200}$ = $\cancel{\frac{1}{12}}$ $\frac{1}{120}$ Sec =

$\frac{1}{2*120} * 10^3$ m Sec = 4.167 msec

transfer time = $\frac{\text{Amount transfered}}{\text{transfer rate}}$ = $\frac{512}{\underbrace{\cancel{4} * 2^{10} * 2^{10}}_{\text{to transfer from}}}$ = $1.22 * 10^{-4}$ byte/sec $\cancel{\text{sec}}$

byte /msec

= $1.22 * 10^{-4} * 10^3$ = 0.122 byte/sec

Controller over head = 6.5 m Sec

$\boxed{1}$ Average time to Read a single sector

= 4.167 + 0.122 + 6.5 + 14 = 24.789 m Sec

2) to Read 16 KB

only transfer Rate changes

transfer Rate = $\left(\frac{16 * 1024}{4 * 2^{10} * 2^{10}}\right) 10^3$ = 3.9 m Sec

Average time to Read 16Kb = 14 + 4.167 + 6.5 + 3.9 = 28.567 msec

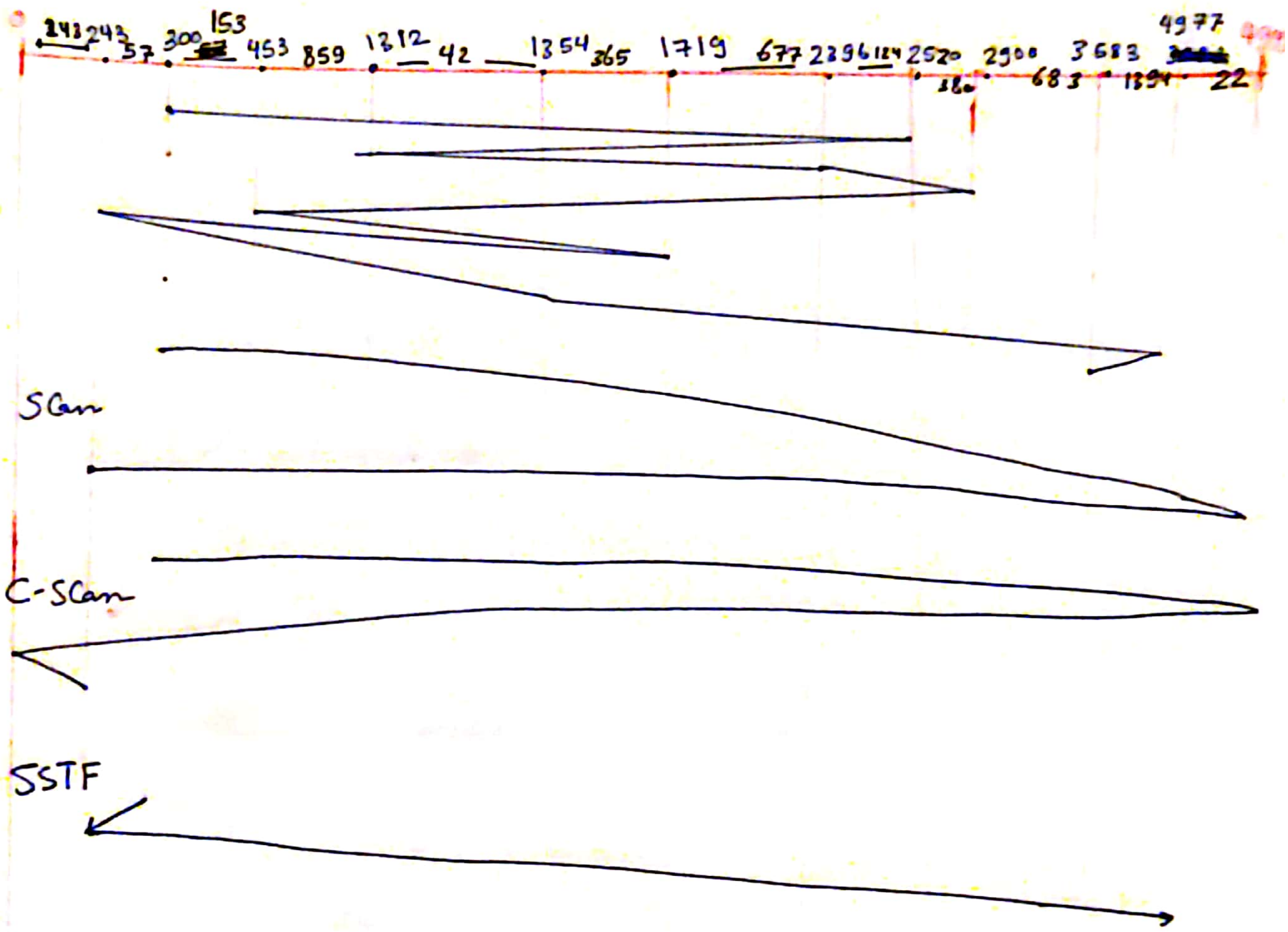
to transfer from MB to byte

$* 1024^2 = 2^{20}$

3) b) 2019/2020 final

Previous Request = 250
Current Request = 300

order = 2520, 1312, 2396, 2900, 453, 1719, 243, 1354, 4977, 3583



1 FCFS - 16333
2 Scan = 9455
3 C-Scan = 9942
4 SSTF = 4791

Final 2018

c)

Proc	Arriving	Burst	priority
P_1	0	6	2
P_2	3	4	1
P_3	5	1	1
P_4	7	6	0
P_5	9	2	1

Find Average waiting and Average Response time + Draw Gantt chart

1 FCFS

P_1	P_2	P_3	P_4	P_5
0 – 6	6 – 10	10 – 11	11 – 17	17 – 19

Average waiting = Average Response , Waiting = All waiting − Arrival

$P_1 = 0-0=0$, $P_2 = 6-3=3$, $P_3 = 10-5=5$, $P_4 = 11-7=4$,

$P_5 = 17-9=8$

Average waiting = Average Response = $\frac{3+5+4+8}{5}$ = 4 units

2 Nonpreemptive priority with FCFS

P_1	P_2	~~P_3~~ P_4	P_3	P_5
0 – 6	6 – 10	10 – ~~16~~ 16	16 – 17	17 – 19

Average waiting = Average Response

Waiting for : $P_1 = 0$ $P_2 = 6-3=3$, $P_3 = 16-5=11$

$P_4 = 10-7=3$, $P_5 = 17-9=8$

Average waiting = Average Response = $\frac{6+11+8}{\cancel{5}\,5}$ = 5 units

3 Non preemptive priority with SJB

P_1	P_3	P_4	P_5	P_2
0 – 6	6 – 7	7 – 13	13 – 15	15 – 19

waiting time for $P_1 = 0$, $P_2 = 15-3=12$, $P_3 = 6-5=1$, $P_4 = 7-7=0$

, $P_5 = 13-9=4$

Average waiting = Average Response = 3.4 units

4) preemptive priority with SRTF

P_1	P_2	P_3	P_2	P_4	P_2	P_5	P_1
0 – 3	3 – 5	5 – 6	6 – 7	7 – 13	13 – 14	14 – 16	16 – 19

Waiting time = All waiting - no of units processed - Arrival time

$P_1 = 16 - 3 - 0 = 13$, $P_2 = 13 - 3 - 3 = 7$, $P_3 = 5 - 0 - 5 = 0$

$P_4 = 7 - 7 = 0$, $P_5 = 14 - 9 = 5$

Average waiting = $\frac{25}{5}$ = 5 units

Average Response = ?

Response for $P_1 = 0$ $P_2 = 3 - 3 = 0$ $P_3 = 5 - 5 = 0$ $P_4 = 7 - 7 = 0$

& $P_5 = 14 - 9 = 5$ Average Response = $5/5$ = 1 unit

Final 2018
Question 4)

magnetic disk with the following parameters

Average seek time	12 MS
Rotation Rate	3600 RPM
transfer Rate	3.5 MByte/sec
Controller overhead	5.5 ms
Number of sector per track	64
Sector size	512 Bytes

1 Average time to Read a single sector

Average time = Average seek time + Average latency + Transfer time + Controlled overhead

$latency = \frac{60}{RPM} = \frac{60}{3600} = \frac{1}{60}$, Average latency = ½ latency

Average latency $= \frac{1}{60} * \frac{1}{2} = \frac{1}{120}$ sec → needed in m sec

Average latency $= \frac{1}{120} * 10^3 = 8.3$ m sec →

transfer time $= \frac{\text{Amount to transfer}}{\text{transfer rate}} = \frac{512}{3.5 * 1024 * 1024 * 10^{-3}} = 0.139$ m Sec

∴ Average time to Read 1 sector = 8.3 + 0.139 + 5.5 + 12 = 25.939 m Sec

2 Average time to Read 8 KB

Only transfer ~~time~~ time changes → لا شي كل من الي صناه فوق ,

transfer time $= \frac{8\,KB}{3.5 * 1024 * 10^{-3}} = 2.23$ m Sec

∴ Average time to Read 8 KB = 12 + 8.3 + 2.23 + 5.5 = 28.03 m Sec

Scanned with CamScanner

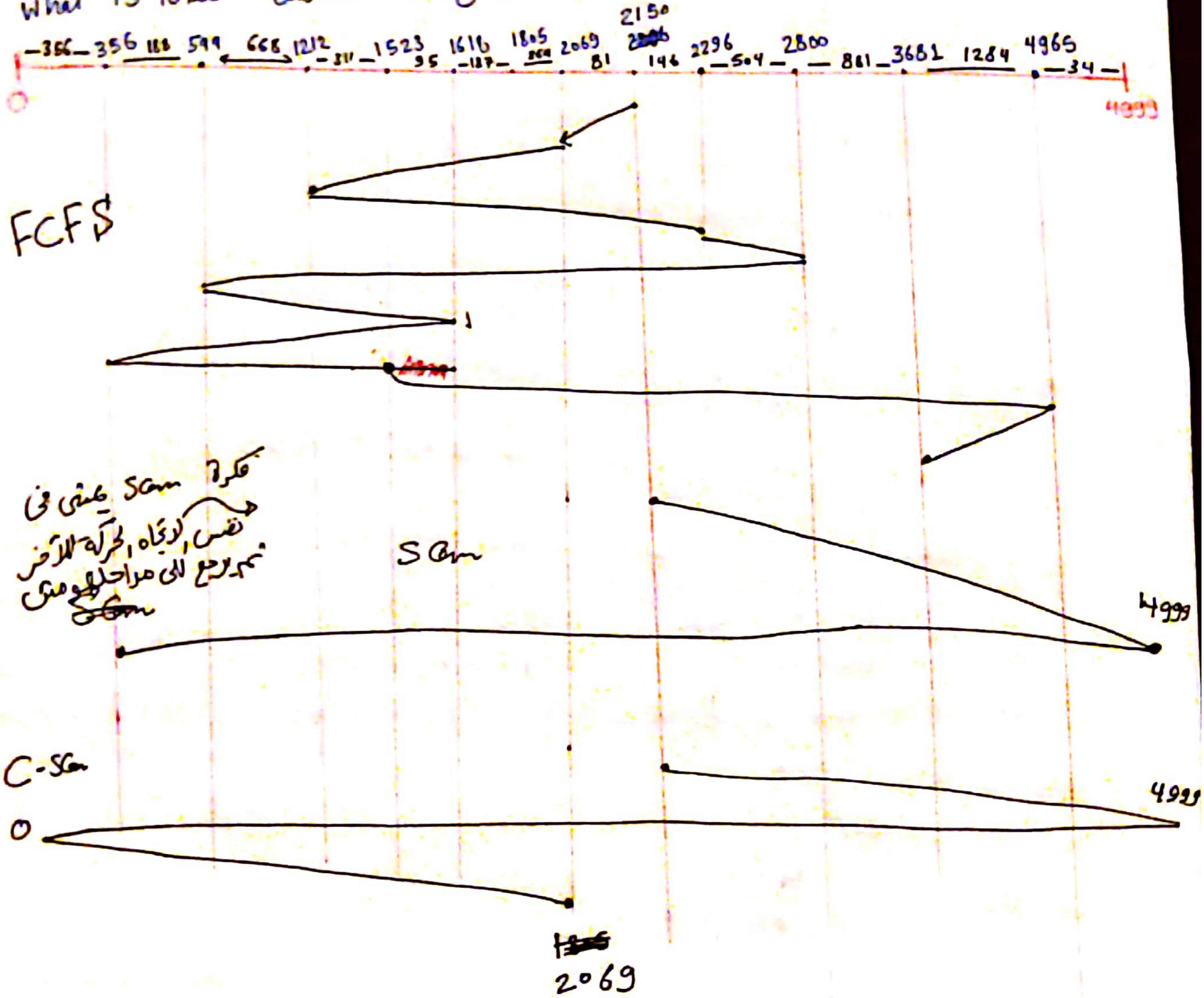
c) 2018 Final

previous Request 1805, Current head = 2150

order is 2069, 1212, 2296, 2800, 544, 1618, 356, 1523, ~~4965~~ 4965, 3681

[1] FCFS [2] SCan [3] C-SCan

what is total distance in cylinders



FCFS = 13011

Scan = 7492

C-Scan = ~~[illegible]~~ 9918